Generative Adversarial Networks with contemporary advances and applications

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StackGAN

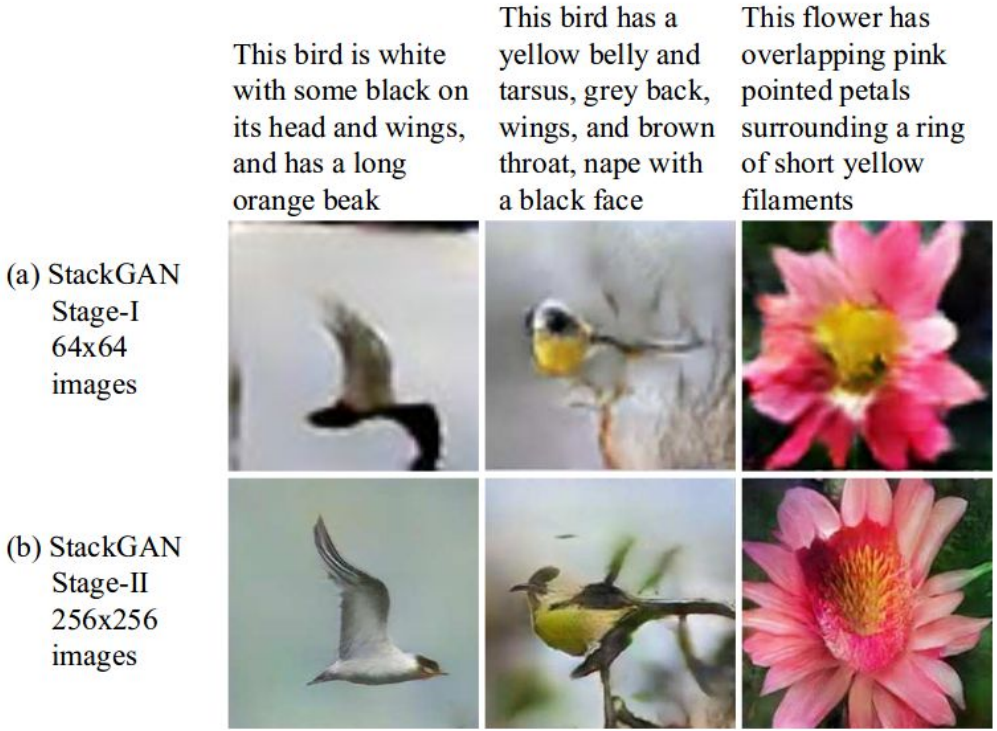
Zhang, H., Xu, T., Li, H., Zhang, S., Wang, X., Huang, X. and Metaxas, D.N., 2017. Stackgan: Text to photo-realistic image synthesis with stacked generative adversarial networks. In *Proceedings of the IEEE international conference on computer vision* (pp. 5907-5915).

2-Stage Network

- Stage 1.
  - Generates 64x64 images
  - Structural information
  - Low detail
- Stage 2.
  - Requires Stage 1 output
  - Upsamples to 256x256
  - Higher detail, photorealistic

Both stages take in the same

conditioned textual input



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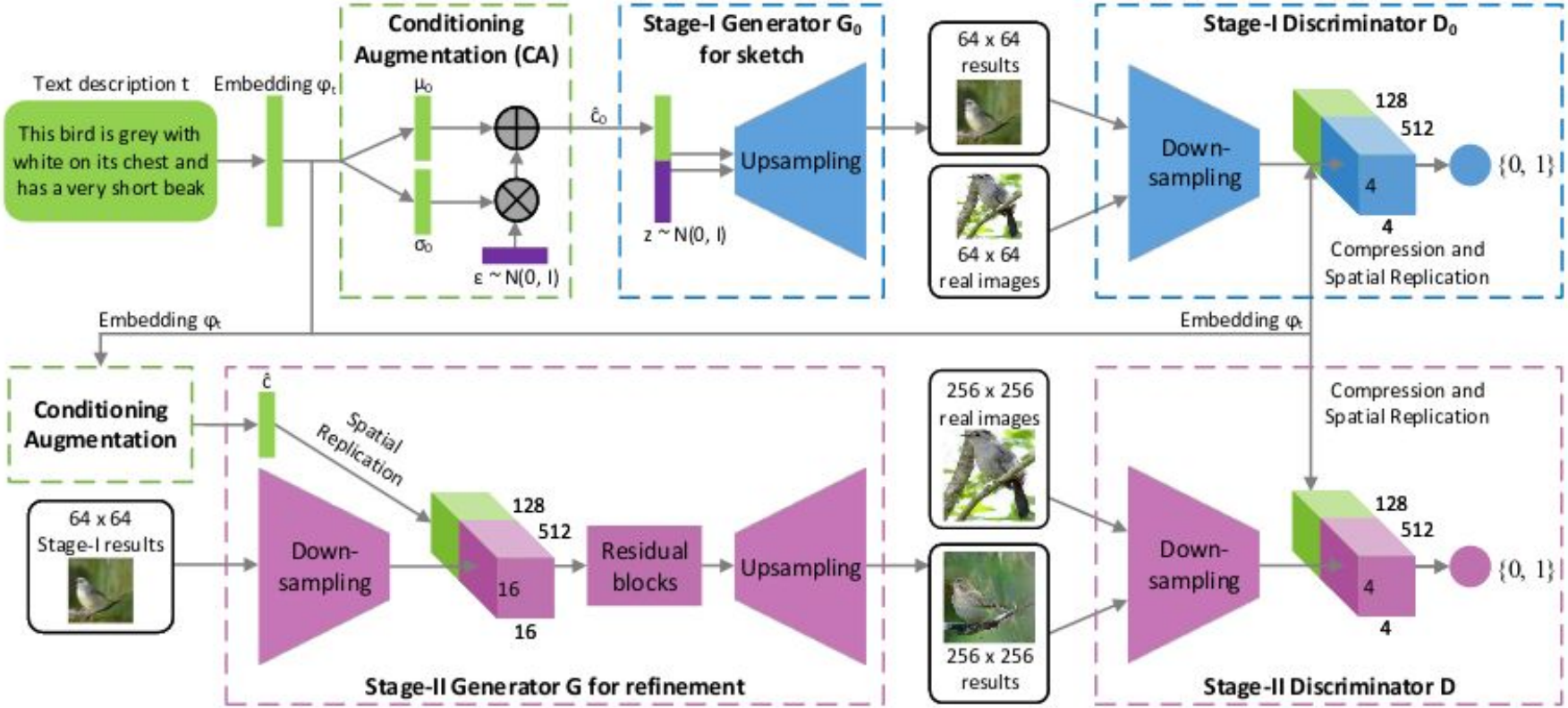
Network Architecture



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Loss Functions (1st Stage)

$$\mathcal{L}_{D_0} = \mathbb{E}_{(I_0,t)\sim p_{data}}[\log D_0(I_0, \varphi_t)] + \mathbb{E}_{z\sim p_z, t\sim p_{data}}[\log(1 - D_0(G_0(z, \hat{c}_0), \varphi_t))]$$

$$\mathcal{L}_{G_0} = \mathbb{E}_{z\sim p_z, t\sim p_{data}}[\log(1 - D_0(G_0(z, \hat{c}_0), \varphi_t))] + \lambda D_{KL}(\mathcal{N}(\mu_0(\varphi_t), \Sigma_0(\varphi_t)) \,||\, \mathcal{N}(0, I)),$$

- Let φ_t be the text embedding of the given description.
- The Gaussian conditioning variables c_0 for text embedding are sampled from $N(\mu_0(\varphi_t), \Sigma_0(\varphi_t))$.
- Real image I_0 and the text description t are from the true data distribution p_{data}.
- Z is a noise vector randomly sampled from a given distribution p_z
- λ is a regularization parameter

Loss Functions (2nd Stage)

$$\mathcal{L}_D = \mathbb{E}_{(I,t)\sim p_{data}}[\log D(I, \varphi_t)] + \mathbb{E}_{s_0\sim p_{G_0}, t\sim p_{data}}[\log(1 - D(G(s_0, \hat{c}), \varphi_t))]$$

$$\mathcal{L}_G = \mathbb{E}_{s_0\sim p_{G_0}, t\sim p_{data}}[\log(1 - D(G(s_0, \hat{c}), \varphi_t))] + \lambda D_{KL}(\mathcal{N}(\mu(\varphi_t), \Sigma(\varphi_t)) \,||\, \mathcal{N}(0, I)),$$

Results

Zhang, H., Xu, T., Li, H., Zhang, S., Wang, X., Huang, X. and Metaxas, D.N., 2017. Stackgan: Text to photo-realistic image synthesis with stacked generative adversarial networks. In *Proceedings of the IEEE international conference on computer vision* (pp. 5907-5915).

StyleGAN

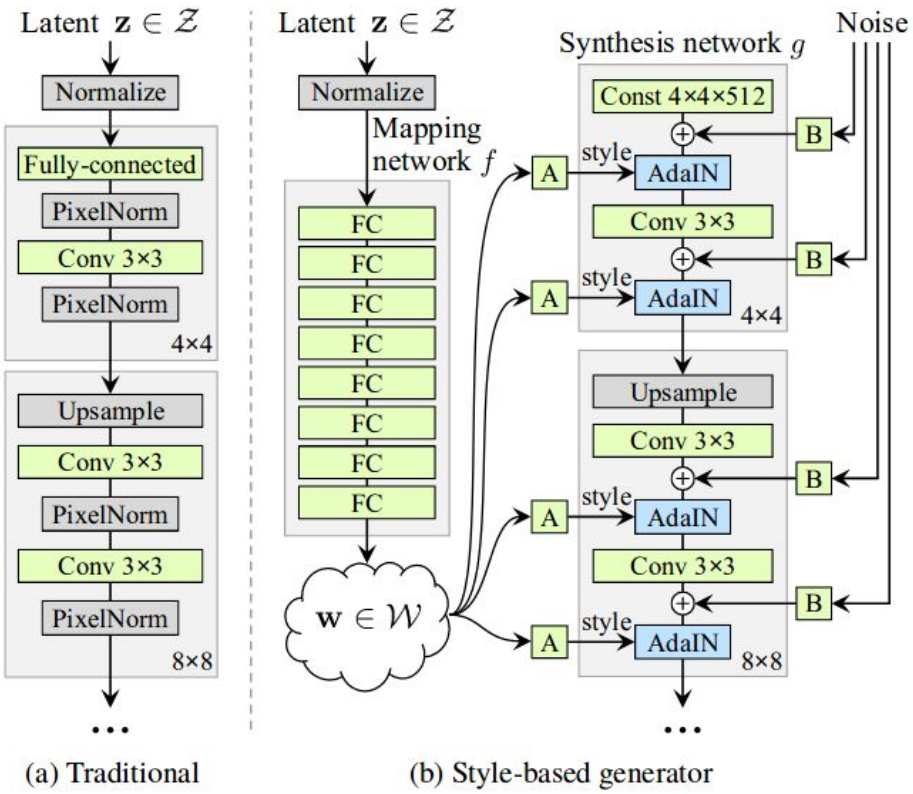
Karras, T., Laine, S. and Aila, T., 2019. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 4401-4410).

Architecture

- Style variations:
  - Affine variations of features
  - Random fluctuations

- The AdaIN operation is defined as

$$\mathrm{AdaIN}(\mathbf{x}_i, \mathbf{y}) = \mathbf{y}_{s,i} \frac{\mathbf{x}_i - \mu(\mathbf{x}_i)}{\sigma(\mathbf{x}_i)} + \mathbf{y}_{b,i},$$



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Results

Uncurated set of images produced by style-based generator with the FFHQ dataset.

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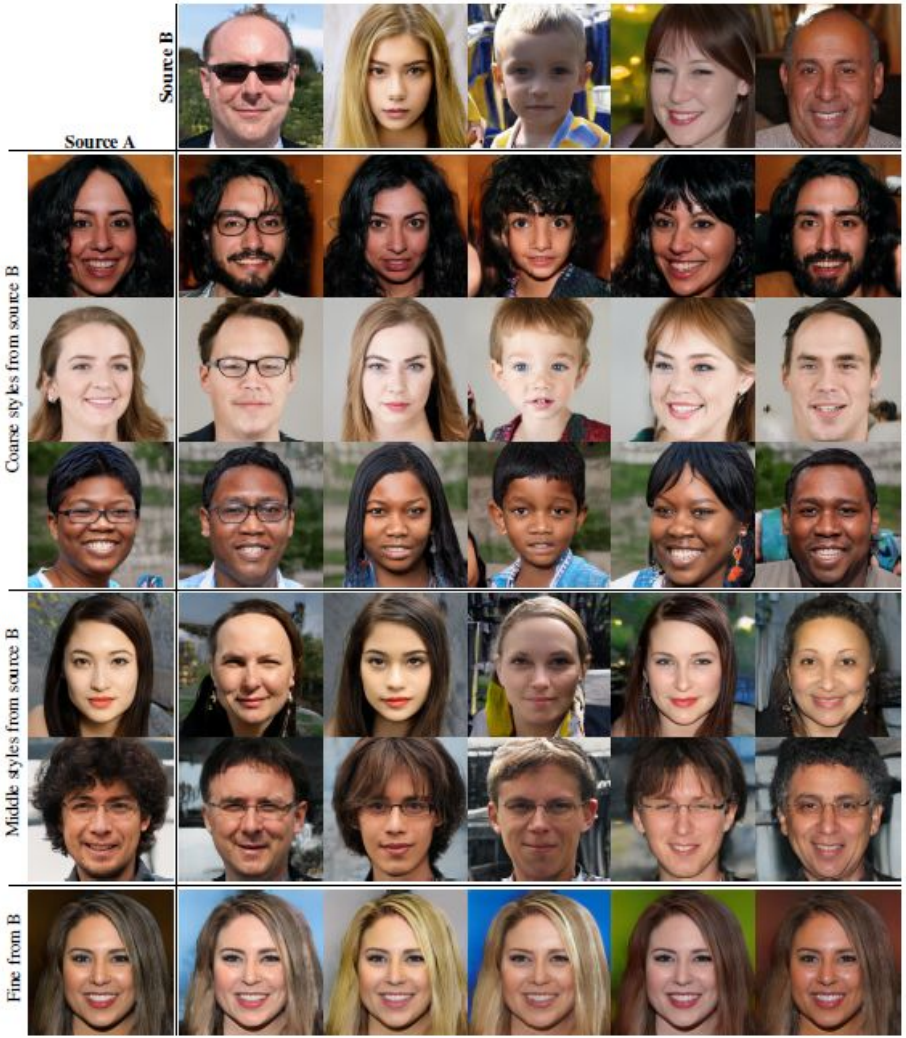
Style mixing

- Generate two latent codes z1, z2 and their transformed versions w1, w2
- Switch from one code w1 to the other code w2 at a random point in the process
- Applying such crossovers at different resolutions / scales

Style mixing

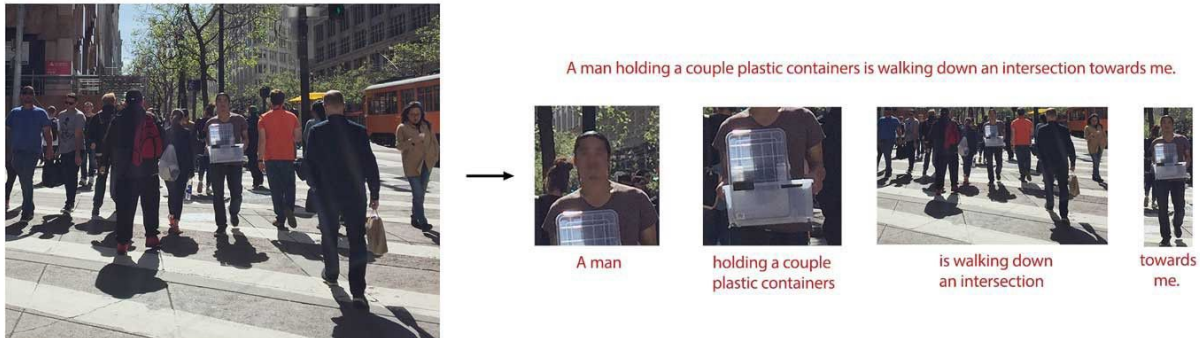


Self-Attention GAN

Zhang, H., Goodfellow, I., Metaxas, D. and Odena, A., 2019, May. Self-attention generative adversarial networks. In *International conference on machine learning* (pp. 7354-7363). PMLR

Introduction

- Self-Attention Generative Adversarial Networks are convolutional neural networks that use the self-attention paradigm to capture long-range spatial relationships in existing images to better synthesize new images.



The highlighted area is the attention area where the network focuses on in generating the specific

https://jonathan-hui.medium.com/gan-self-attention-generative-adversarial-networks-sagan-923fccde790c

Motivation

- Most GAN-based models for image generation are built using convolutional layers.

- Convolution processes the information in a local neighborhood, thus using convolutional layers alone is computationally inefficient for modeling long-range dependencies in images.

- The motivation behind the SA-GAN is enabling both the generator and the discriminator to efficiently model relationships between widely separated spatial region.

Results

- The proposed SAGAN generates images by leveraging complementary features in distant portions of the image rather than local regions of fixed shape to generate consistent objects/scenarios.
- In each row, the first image shows five representative query locations with color coded dots.
- The other five images are attention maps for those query locations, with corresponding color coded arrows summarizing the most-attended regions

Attention map contd.

$$\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{W_g}\boldsymbol{x} \qquad \boldsymbol{x} \in \mathbb{R}^{C\times N}, \boldsymbol{W_f} \in \mathbb{R}^{\bar{C}\times C}, \bar{C} = C/8$$

$$\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{W_f}\boldsymbol{x} \qquad \boldsymbol{W_g} \in \mathbb{R}^{\bar{C}\times C}$$

$$s_{ij} = \boldsymbol{f}(\boldsymbol{x_i})^T \boldsymbol{g}(\boldsymbol{x_j})$$

$$\beta_{j,i} = \frac{\exp(s_{ij})}{\sum_{i=1}^{N} \exp(s_{ij})} \qquad \beta \in \mathbb{R}^{N\times N}$$

$\beta_{j,i}$ indicates the extent to which the model attends to the ith location when synthesizing the jth region.

- For each spatial location, an attention map is created which acts as a mask.
- *βij* is interpreted as the impact of location *i* when rendering the location j.

Attention output

$$\boldsymbol{h}(\boldsymbol{x_i}) = \boldsymbol{W_h}\boldsymbol{x_i} \qquad \boldsymbol{W_h} \in \mathbb{R}^{C\times C}$$

$$\boldsymbol{o_j} = \sum_{i=1}^{N} \beta_{j,i}\boldsymbol{h}(\boldsymbol{x_i}) \qquad \boldsymbol{o} \in \mathbb{R}^{C\times N}$$

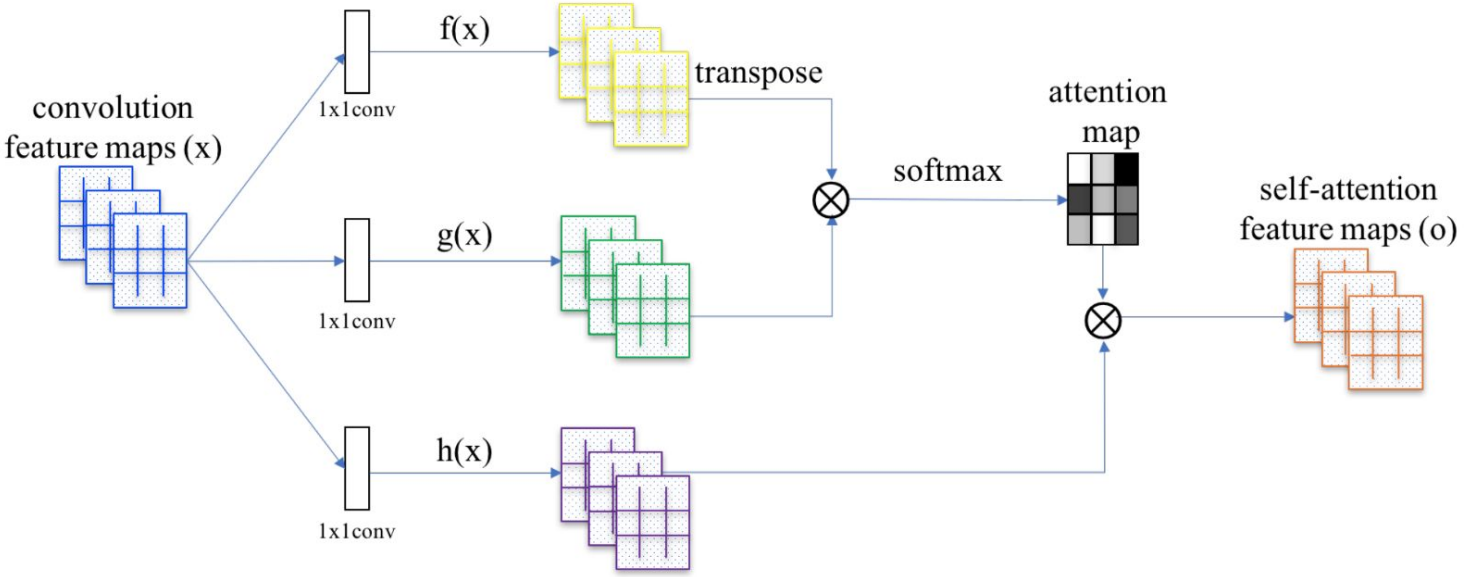
The final output of this convolutional layer is:

$$\boldsymbol{y_i} = \gamma \boldsymbol{o_i} + \boldsymbol{x_i}$$

where γ is initialized as 0 so the model will explore the local spatial information first before refining it with self-attention.

Attention output

- Multiply *x* with W_h (trainable model parameters) and merge it with the attention map β to generate the self-attention feature map output o.



Results

SN on G/D+TTUR (10k, FID=99.04) | SN on G/D+TTUR (160k, FID=40.96) | SN on G/D+TTUR (260k, FID=34.62) | SN on G/D+TTUR (1M, FID=22.96)

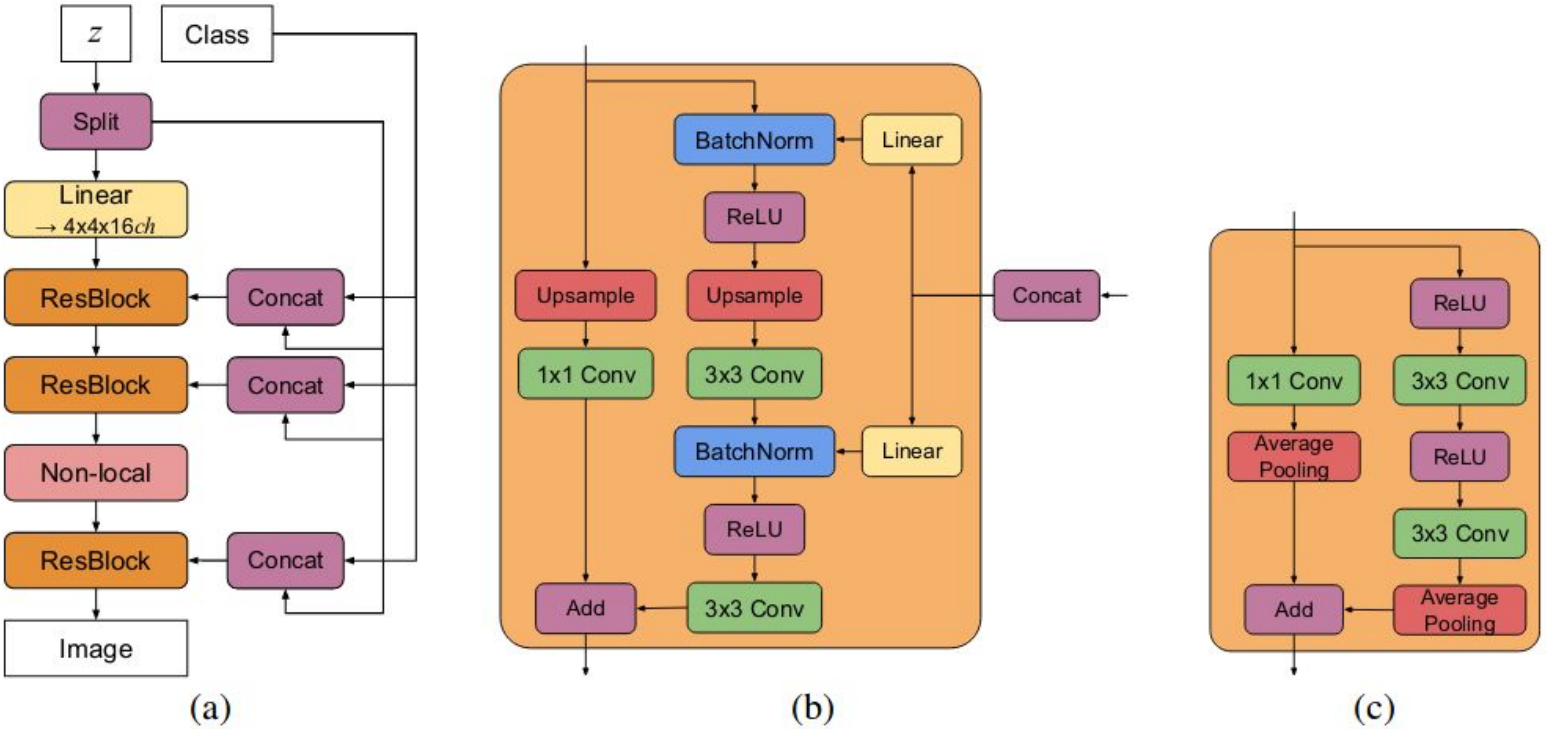
128×128 examples randomly generated by the SAGAN models

BigGAN

Large Scale GAN Training for High Fidelity Natural Image Synthesis

Brock, A., Donahue, J. and Simonyan, K., 2018. Large scale GAN training for high fidelity natural image synthesis. *arXiv preprint arXiv:1809.11096*.

Architecture



a) **A typical architectural layout for BigGAN's G;**
b) **A Residual Block (ResBlock up) in BigGAN's G.**
c) **A Residual Block (ResBlock down) inBigGAN's D.**

Techniques

- Increase batch size by a factor of 8
- Increase channel width
- Increase depth

- Shared class embeddings for conditional batch norm
- Skip connections to multiple layers of generator

- Truncation trick: Truncate the range of the input latent codes when sampling from the distribution

- Orthogonal regularization

$$R_\beta(W) = \beta \parallel W^T W - I \parallel_F^2$$

Truncation Trick

- Effect of restricting the range of the latent variable
- Balancing between variability and fidelity

Results

Samples generated by BigGAN model at 256×256 resolution

Results

Samples generated by BigGAN model at 512×512 resolution